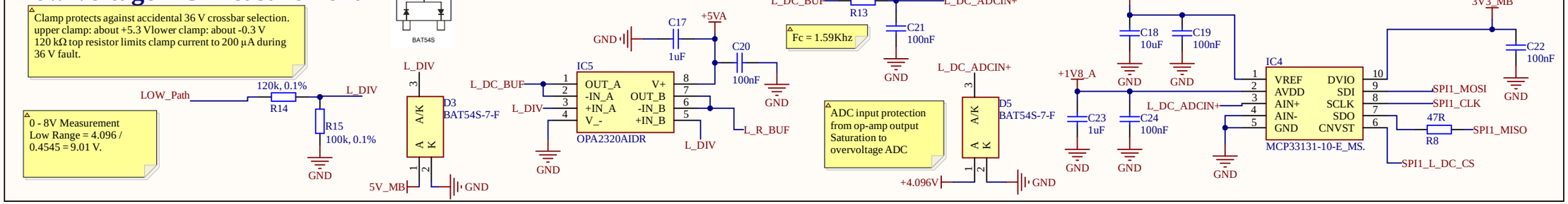
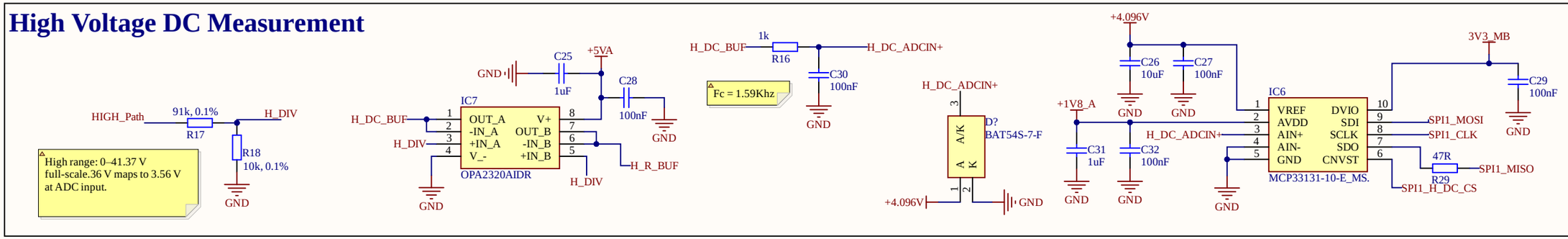
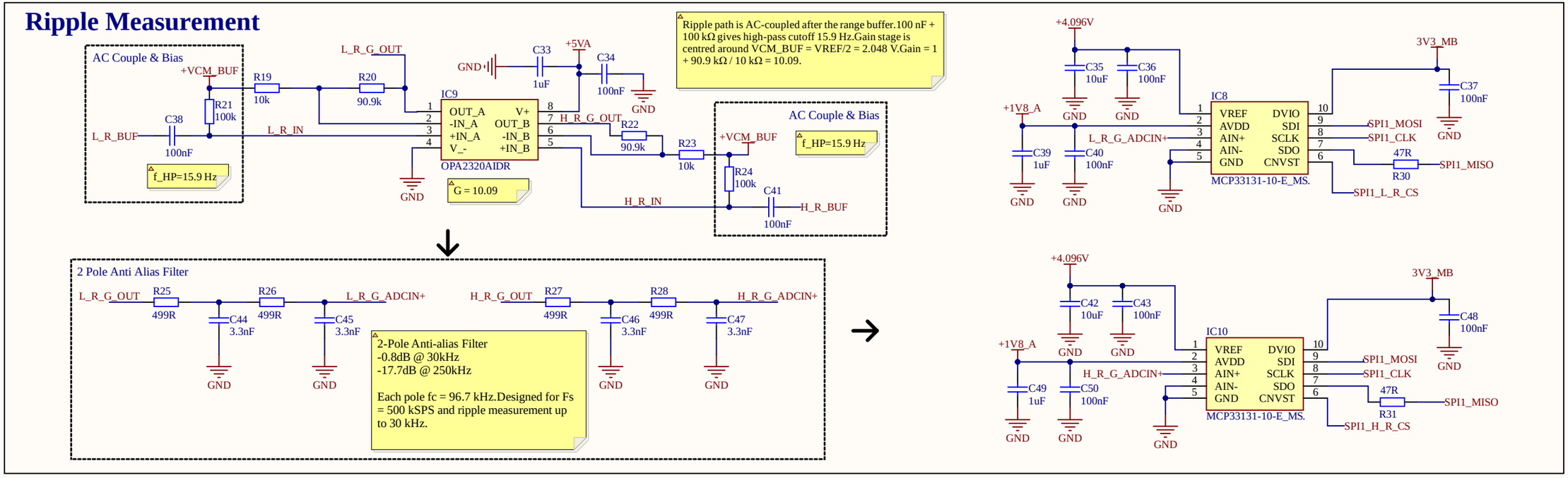
Low Voltage DC Measurement



High Voltage DC Measurement

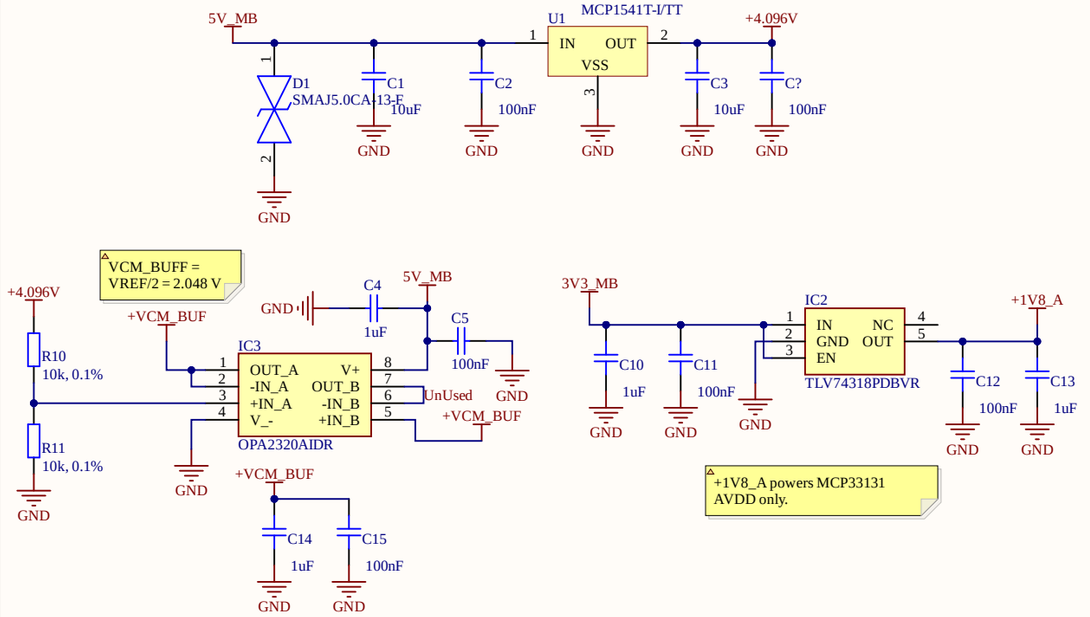


Ripple Measurement

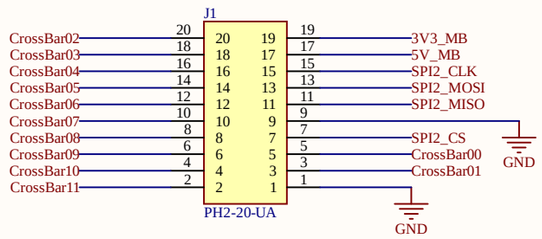


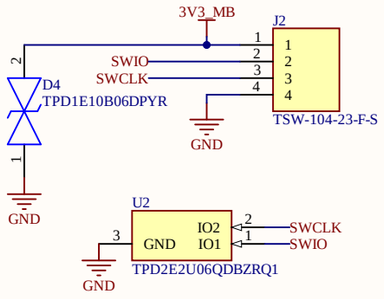
Title			
Size: B	Number:	Revision:	
Date: 7/23/2026		Sheet of	
File: AnalogPaths.SchDoc		Drawn By:	

Reference Voltages

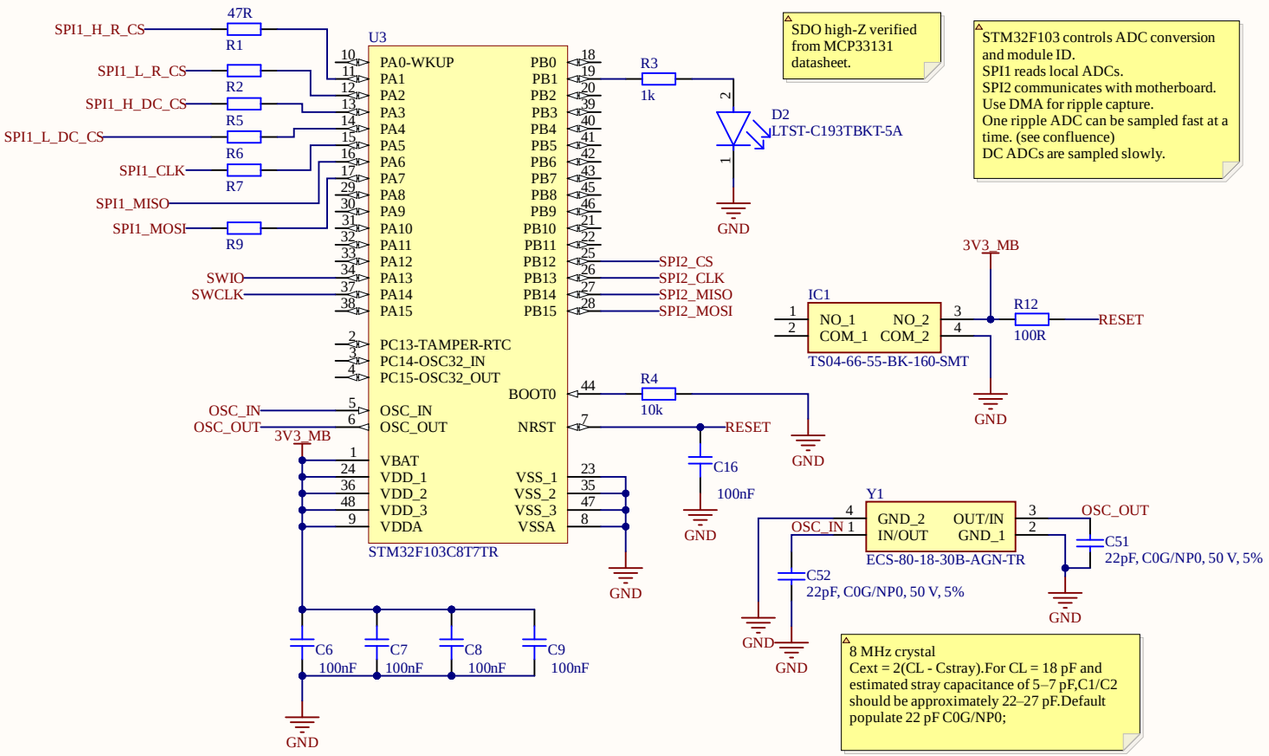


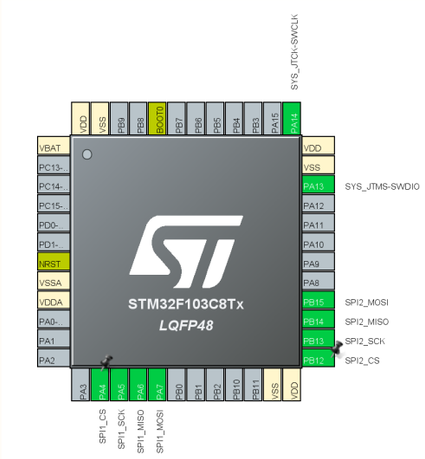
Connectors





MCU





Calibration Test Points:
TP_GND
TP_4V096_REF
TP_VCM_BUF
TP_1V8_A
TP_LOW_DIV
TP_HIGH_DIV
TP_LOW_DC_ADCIN
TP_HIGH_DC_ADCIN
TP_LOW_RIPPLE_ADCIN
TP_HIGH_RIPPLE_ADCIN

Each measurement path is calibrated in firmware using known external input voltages.

DC paths use two-point calibration: V = mCode + b.

Ripple paths use known AC input amplitude and mean-subtracted waveform capture.

MCP1541 initial tolerance is corrected by calibration

Title		
Size: A3	Number	Revision
Date: 7/23/2026		Sheet of
File: PowerAndMCU.SchDoc		Drawn By: